

Organic Chemistry Quick Revision

Organic chemistry deals with **organic molecules**, which all contain **carbon atoms**. These molecules are mostly formed from living organisms, but can also be made synthetically.

Representing Organic Molecules

To represent organic molecules, you can use:

- **Empirical Formula**: The simplest whole number ratio of each element in a compound.
- **Molecular Formula**: Gives the exact number of each element within the compound.
- **General Formula**: A type of empirical formula that represents the composition of any member of an entire class of compounds.
- **Structural Formula**: Shows the arrangement of the atoms in the molecule.
- **Displayed Formula**: Gives the symbol of all the different elements and their bonding.

Homologous Series

A **homologous series** is a number of compounds which have the same general formula and the same functional group.

- We can use the general formula to work out the molecular formula of any member of the homologous series.
- They differ by CH_2 in their molecular formula from neighboring compounds.
- They show gradual variation in their physical properties, such as melting point and boiling point.
- Because they have the same functional groups, they share the same chemical properties or same reactions.

The main homologous series we will learn about are **hydrocarbons**. These include:

- **Saturated Hydrocarbons:** Represented by **alkanes**, which only have single covalent bonds between carbon atoms.
- **Unsaturated Hydrocarbons:** Represented by **alkenes**, which have a double bond.
- **Alcohols:** Have the functional group of OH .
- **Carboxylic Acids:** Have the functional group of a carboxy group ($C = O$).
- **Esters:** Have the functional group $C = O$.

Naming Organic Molecules

To name different organic molecules, we use a **prefix** and a **suffix**.

- **Prefix:** Refers to the number of carbon atoms in the chain or molecule.

Number of Carbon Atoms	Prefix
1	Meth-
2	Eth-
3	Prop-
4	But-
5	Pent-
6	Hex-

- **Suffix:** Indicates the homologous series.

Homologous Series	Suffix	Example
Alkanes	-ane	Propane
Alkenes	-ene	Propene
Alcohols	-ol	Ethanol
Carboxylic Acids	-oic acid	Ethanoic acid
Esters	-yl, -anoate	Methyl ethanoate

- For alcohols, remove the "-e" from the alkane name and replace it with "-ol".
- For carboxylic acids, remove the "-e" from the alkane name and replace it with "-oic acid".
- For esters, the first part comes from the alcohol (alkyl group) and the second part comes from the carboxylic acid (-anoate).

Isomers

In organic molecules, two or more different molecules can share the same molecular formula but be different compounds. These are called **isomers**. There are three types:

- **Chain Isomers:** Compounds with the same molecular formula but different carbon chain connections.
 - Example: C_4H_{10} can be either butane or methylpropane.
 - They have the same chemical properties but different physical properties.
- **Position Isomers:** Compounds with the same molecular formula and functional group, but the functional group is at a different position on the carbon chain.
 - Example: C_4H_8 can be but-1-ene or but-2-ene.
 - They have the same chemical properties but different physical properties.
- **Functional Group Isomers:** Compounds with the same molecular formula but different functional groups.
 - Example: $C_4H_8O_2$ can be butanoic acid or methyl propanoate.
 - They have different chemical properties and different physical properties.
 - All types of isomers have different physical properties, but only functional group isomers have different chemical properties.

Hydrocarbons

Hydrocarbons contain only carbon and hydrogen.

- They can come in varying lengths and can be branched.
- Many are found in living organisms and their fossils.
- Others can be prepared in the laboratory.
- We use hydrocarbons in everyday life as fuel (natural gas, acetylene, propane, butane, etc.) and in the petrochemical industry (plastics).

Alkanes

Alkanes are saturated hydrocarbons that only contain single covalent bonds between carbon atoms.

- Each carbon atom can form four different bonds with carbon or hydrogen atoms.
- The general formula for alkanes is C_nH_{2n+2} .
- To name alkanes, use the prefix that refers to the number of carbon atoms and the suffix "-ane".

Name	Molecular Formula
Methane	CH_4
Ethane	C_2H_6
Propane	C_3H_8
Butane	C_4H_{10}

Chemical Bonding in Alkanes

When drawing any compound, each **carbon atom** must have four bonds. For **methane**, all bonds are with hydrogen. For **ethane**, there is one bond with another carbon atom and the rest with hydrogen.

Physical Properties: Boiling Point

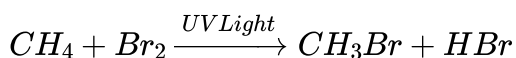
As the number of carbon atoms in an alkane chain increases, the **boiling point** of the compound also increases. This is due to the increase in **intermolecular forces** between the compounds.

Higher intermolecular forces require more energy to break the bonds between the compounds.

Reactions of Alkanes

Alkanes are fairly **inert** due to their single bonds (either carbon-hydrogen or carbon-carbon). They don't tend to react easily and are generally unreactive.

The primary reaction for alkanes is with **halogens** like chlorine and bromine, which only occurs in the presence of **UV light**. For instance, methane reacts with bromine under UV light in a **substitution reaction**:

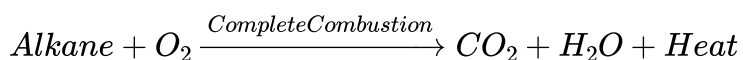


In this reaction, one hydrogen atom from the alkane is substituted with one bromine atom, resulting in **bromomethane** and **hydrogen bromide**.

Combustion of Alkanes

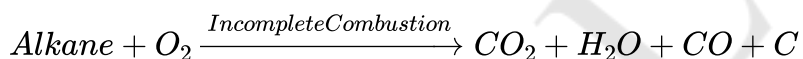
Alkanes (hydrocarbons) are mainly used as **fuel** because they undergo **complete combustion** in the presence of excess oxygen.

During complete combustion, alkanes are fully oxidized to produce **carbon dioxide** and **water**:



This reaction is **exothermic**, releasing a lot of heat, which is why alkanes are a good source of energy.

If there isn't enough oxygen, alkanes can undergo **incomplete combustion**, producing carbon monoxide and carbon (soot) in addition to carbon dioxide and water:



Carbon monoxide (CO) is a colorless, odorless, and silent killer because it binds to hemoglobin in red blood cells with a much higher affinity than oxygen. This prevents oxygen transport and leads to **suffocation**.

Acid Rain and Hydrocarbons

The high temperatures in car engines can cause nitrogen and oxygen from the air to react, producing **nitrogen oxides** like nitric oxide (NO) and nitrogen dioxide (NO₂). Additionally, sulfur present in hydrocarbons can react with oxygen during combustion to produce **sulfur dioxide (SO₂)**.

Acid rain is formed when these oxides dissolve in rainwater, creating acids such as **nitric acid** and **sulfuric acid**.

Problems with Acid Rain

- Corrosion of rocks and buildings
- Altering the pH of soil and rivers, affecting ecosystems
- Corrosion of limestone, damaging buildings and statues

Formation and Fractional Distillation of Crude Oil

Crude oil is formed from the remnants of living organisms that lived millions of years ago. These remains have been buried under soil for a long time, subjected to high pressure and temperature, transforming them into crude oil.

Crude oil is a mixture of many different hydrocarbons that must be separated before use. This is done through **fractional distillation** using a furnace and a fractionating column.

Process of Fractional Distillation

1. Crude oil is heated in a furnace to a high temperature.
2. The different components boil and enter the fractionating column.
3. The column has a temperature gradient, with the highest temperature at the bottom and decreasing temperatures as you go up.
4. The oil evaporates and condenses at different levels based on their boiling points.

Fractions Separated

Fraction	Boiling Point	Use
Gases (Butane/Propane)	Low	Fuel
Petrol	Low-Medium	Gasoline
Kerosene	Medium	Jet fuel
Diesel	Medium-High	Diesel fuel
Fuel Oil	High	Industrial fuel
Lubricating Oils/Wax/Asphalt	Very High	Lubricants, paraffin, road paving

Cracking of Hydrocarbons

Cracking is used to break down long-chain hydrocarbons into shorter, more useful ones.

The long-chain hydrocarbons are heated until vaporized and then broken down using one of two methods:

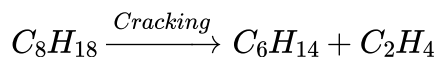
Catalytic Cracking

Vaporized hydrocarbons are passed over a hot catalyst, such as silica or alumina, breaking them down into shorter chains.

Steam Cracking

Hydrocarbon vapors are mixed with steam and heated to a very high temperature (600-700°C), causing thermal decomposition.

Cracking Equation Example



(Octane → Hexane + Ethene)

Alkenes

Alkenes are unsaturated hydrocarbons with a double bond between two carbon atoms.

Alkenes are unsaturated hydrocarbons with the general formula C_nH_{2n} .

The simplest alkene is **ethene** (ethylene), which has two carbon atoms and a double bond. The functional group for alkenes is the carbon-carbon double bond.

Alkenes

Alkenes are hydrocarbons with one or more carbon-carbon double bonds.

- Naming alkenes:
 - Prefix: number of carbon atoms
 - Suffix: "-ene"

Here are the first four members of alkenes:

Alkene	# of Carbon Atoms	Formula	Notes
Ethene	2	$CH_2 = CH_2$	Cannot have a double bond with only one carbon atom.
Propene	3	$CH_3 - CH = CH_2$	
1-Butene	4	$CH_3 - CH_2 - CH = CH_2$	Isomers (position isomers): same molecular formula, but the position of the double bond is different. The double bond is between the second and third carbon atoms, rather than the first and second.
2-Butene	4	$CH_3 - CH = CH - CH_3$	

Addition Reaction of Alkenes

A main reaction of alkenes is the **addition reaction**. This is when the double bond of an alkene is broken, and two different atoms are added.

- Alkenes react with bromine to give 1,2-dibromoalkanes.
- Alkenes react with chlorine to give 1,2-dichloroalkanes.

Bromine Water Test

Bromine water is used to test for the presence of a double bond.

- Bromine water is orangey in color.
- If you add bromine water to a sample with a double bond, the color will disappear. This is because the bromine has reacted with the double bond to give you the 1,2-dibromoalkane.

Alcohols

Alcohols contain the functional group OH.

- Naming alcohols:
 - Prefix: number of carbon atoms
 - Suffix: "-ol"

Here are some alcohols:

Alcohol	# of Carbon Atoms	Formula	Notes
Methanol	1	CH_3OH	Found in beer, wine, and alcoholic drinks.
Ethanol	2	CH_3CH_2OH	
Propanol	3	$CH_3CH_2CH_2OH$	
Butanol	4	$CH_3CH_2CH_2CH_2OH$	

Ethanol can be used as a fuel for vehicles.

Production of Ethanol

1. From ethene and steam, using acid as a catalyst (addition reaction):

- Ethene + Water → Ethanol
 - $C_2H_4 + H_2O \Rightarrow C_2H_5OH$
- Acid (sulfuric or phosphoric acid) catalyst
- High temperature (300°C)
- High pressure (60-70 atm)

Ethene is produced from crude oil via cracking. The addition of water (steam) breaks the double bond and adds an *OH* group to one carbon and a hydrogen to the other, forming ethanol.

- Advantages: cheap reaction, no waste produced
- Disadvantages: ethene is produced from crude oil, which is a nonrenewable source of energy

2. Fermentation of sugar (e.g., glucose):

- Enzymatic catalytic reaction using yeast
 - Yeast provides the enzyme needed for fermentation
- Optimum temperature: 25-35°C
 - Low temperature: enzyme is inactive
 - High temperature: enzyme (protein) is denatured
- Slightly acidic conditions
- Absence of oxygen (air must be kept out)
 - Ethanol in the presence of oxygen will be oxidized to ethanoic acid

Glucose (sugar) is considered a renewable source for the production of ethanol.

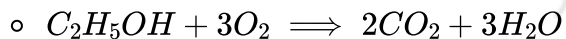
- Advantage: uses glucose or sugars, which are renewable
- Disadvantage: ethanol produced is very diluted and needs to be concentrated and purified

Concentration of Ethanol from Fermentation

1. Filtration: separate yeast cells and insoluble substances
 - Ethanol concentration reaches 15%, the yeast will die
2. Fractional distillation: concentrate the ethanol
 - Boiling point of ethanol is less than the boiling point of water

Reactions of Alcohols

1. Oxidation: alcohols are oxidized to give carboxylic acid.
 - Oxidizing agent: potassium dichromate (VI) in dilute sulfuric acid
 - Ethanol gives ethanoic acid
2. Microbial oxidation: microbes use ethanol as a source of energy and oxidize the ethanol using oxygen from the air. Carboxylic acid is produced as a side product.
3. Complete combustion: alcohols undergo complete combustion in the presence of oxygen to give carbon dioxide and water (used as fuel).



Carboxylic Acids

Carboxylic acids have the carboxy group functional group: COOH , which consists of a carbon with a double bond to an oxygen and a single bond to an OH group.

- Naming carboxylic acids:
 - Prefix: number of carbon atoms (including the carbon in the carboxy group)
 - Suffix: "-oic acid"

Here are the first four members of carboxylic acids:

Carboxylic Acid	# of Carbon Atoms	Formula	Notes
Methanoic acid	1	$HCOOH$	Simplest carboxylic acid
Ethanoic acid	2	CH_3COOH	Found in vinegar
Propanoic acid	3	CH_3CH_2COOH	
Butanoic acid	4	$CH_3CH_2CH_2COOH$	

Properties of Carboxylic Acids

Carboxylic acids are weak acids and have typical acidic properties:

- Dissolve in water to produce an acidic solution with a pH less than 7.
- React with metal carbonates to give a salt, carbon dioxide, and water.
 - $2CH_3COOH + Na_2CO_3 \Rightarrow 2CH_3COONa + CO_2 + H_2O$
 - Carbon dioxide can be recognized as bubbles or effervescence.
- React with more reactive metals to produce salt and hydrogen.
 - $2CH_3COOH + Mg \Rightarrow (CH_3COO)_2Mg + H_2$
 - Hydrogen gas is produced.

Reactions of Carboxylic Acids

Carboxylic acids exhibit typical acid behaviors:

- React with **metals** to form a **salt** and **hydrogen gas**.
 - The reaction is identifiable by observing hydrogen bubbles.
 - Example: Ethanoic acid + Sodium → Sodium ethanoate + Hydrogen gas
- React with **bases** to form **salt and water**.
 - The **carboxylic acid group** (specifically the hydrogen atom) is the source of these acidic properties.

Formation of Carboxylic Acids

Carboxylic acids are formed through the **oxidation of alcohols**. The resulting carboxylic acid will have the same number of carbon atoms as the alcohol.

Alcohol	Carboxylic Acid
Ethanol	Ethanoic acid
1-Propanol	Propanoic acid
1-Butanol	Butanoic acid

Esters

Esters are part of a homologous series with the functional group **COO**.

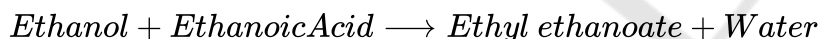
Formation of Esters

Esters are formed through a **condensation reaction** between an alcohol and a carboxylic acid.

Condensation Reaction: a reaction where a water molecule is removed.

During this reaction:

- The **-OH** group is removed from the **carboxylic acid**.
- A **hydrogen atom** is removed from the **alcohol**.
- The **-OH** and **hydrogen atom** combine to form a water molecule.
- The **oxygen** from the alcohol bonds to the **carbon** from the carboxylic acid, forming the ester with the COO functional group.



Naming Esters

To name esters:

1. Start with the name of the alcohol part first (as a prefix).
 - Example: Ethanol becomes "ethyl"
2. Name the carboxylic acid part.
 - Example: Ethanoic acid becomes "ethanoate".
3. Combine the two names.
 - Example: Ethanol + Ethanoic Acid becomes "ethyl ethanoate"